47 Secondary Neoplasia (Other Than PTLPS)

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47.1 Definitions

Secondary neoplasia (SN) after HCT includes any malignant disorder occurring after HCT, irrespectively, if related or not to transplantation. For an individual patient, a clear relationship between HCT and SN often cannot be demonstrated. Posttransplant lymphoproliferative disorders are discussed elsewhere (see Chap. 45).

47.2 Types of Secondary Neoplasia After HCT

	Therapy-related myeloid neoplasms (t-MN) (Pedersen-Bjergaard et al. 2000)	Donor-derived malignancy (DDM) (Sala-Torra et al. 2006; Wiseman 2011; Engel et al. 2019)	Second solid neoplasms (SSN) (Kolb et al. 1999; Rizzo et al. 2009)
Definition	t-MDS or t-AML after exposition chemo- or radiation therapy	Hematologic neoplasms occurring in grafted donor cells	Solid cancers of any site and histology occurring after HCT
Occurrence	Mainly after auto-HCT Not excluded after allo-HCT (Yamasaki et al. 2017)	After allo-HCT only	After allo-HCT and auto-HCT
Appearance	Within the first 10 years mainly	Variable	Increasing incidental rate with longer follow-up
Prognosis	Poor	Poor	Depends mainly on the cancer type

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A. Sureda et al. (eds.), *The EBMT Handbook*, https://doi.org/10.1007/978-3-031-44080-9_47

47.3 Pathophysiology

47.3.1 Therapy-Related Myeloid Neoplasms

t-MN are mainly associated with cytotoxic chemotherapy and radiation therapy that the patient has received either before HCT or as conditioning. The causal role of ionizing radiation in the development of myeloid neoplasms has been demonstrated in atomic bomb survivors of Hiroshima/Nagasaki and in medical radiation workers employed before 1950.

Responsible cytotoxic drugs:

- Alkylating agents, anthracyclines, and topoisomerase II inhibitors.
- To a lesser extent antimetabolites and purine analogs.
- Controversy exists on the role of azathioprine, methotrexate, hydroxyurea, and 6-mercaptopurines used for the treatment of malignant and nonmalignant diseases.

t-MN occur mainly after auto-HCT, where the healthy HCT has been exposed to cytotoxic effect (Diamond et al. 2023). Detecting clonal hematopoiesis mutations in cryopreserved cells before auto-HCT has been associated with a higher risk of t-MN (Gramegna et al. 2022). Rarely t-MN can be observed after allo-HCT. Persistent microchimerism with exposed residual recipient cells or autologous recovery may explain the development of t-MN after allo-HCT (Hoshino et al. 2022). After HCT for sickle cell disease, the 10-year incidence of leukemia and MDS is increased, mainly when conditioned with reduce-intensity and TBI-containing regiment. The onset of t-MN and graft failure after HCT are closely related (Eapen et al. 2023).

Today, increasingly cytotoxic drugs are applied after the allo-HCT, either as GVHD prophylaxis (posttransplant CY) or to prevent disease recurrence (posttransplant maintenance). Posttransplant CY does not seem to increase the risk of t-MN (Majzner et al. 2017; TCT). We do not yet know whether posttransplant maintenance therapy could increase the risk of t-MN after allo-HCT.

A germline cancer predisposition has been demonstrated in 15–20% of t-MNs, and acquired mutagenic effect of cytotoxic therapy with clonal hematopoiesis of indetermined potential (CHIP) is frequently the first step in the multihit development of t-MNs (Voso et al. 2021).

47.3.2 Donor-Derived Malignancy (DDM)

The pathogenesis of donor-derived hematological malignancies is not fully understood but is likely multifactorial (Sala-Torra et al. 2006; Wiseman 2011; Williams et al. 2021; Gibson et al. 2022):

- Treatment damage to bone marrow microenvironment from previous chemotherapy, radiation, and treatments to prevent GVHD
- Transplantation of a malignant clone, or germline or somatic mutations from the donor
- Stress of rapid clonal expansion after transplant

Reported donor cell leukemia are AML, MDS, ALL, CML, and lymphoid neoplasms including CLL (Engel et al. 2019). The risk of DDM in allogeneic hematopoietic cell transplantation is driven by somatic myelodysplastic syndrome-associated mutations or germline predisposition in donors (Gibson et al. 2022).

Clonal hematopoiesis can be transmitted from a donor to a recipient during allo-HCT. Exclusion of candidate donors with clonal hematopoiesis is controversial since its impact on recipient outcomes and graft alloimmune function is uncertain.

Over 20% of donor-derived myeloid neoplasms carry chromosome 7 abnormalities. Gene sequencing of the donor cells allowed to detect a number of candidate genes that could contribute to the development of DDM (Williams et al. 2021). Malignant clones transferred to the recipient can be of lymphoid origin, observed in older donors, and may evolve into a lymphoid neoplasm in the immunosuppressed host.

47.3.3 Second Solid Neoplasms (SSN)

Little is known about pathogenesis of SSN after HCT. An interaction between cytotoxic treatment, genetic predisposition, environmental factors, viral infections, GVHD, and its immunosuppression may play a role.

Three main types of SSN (Rizzo et al. 2009):

- Radiation-related SSN
  - Proven for thyroid, breast, and brain cancers
  - Occur after a long latency (≥10 years after radiation)
  - Is dose related
- GVHD/immunosuppression-related SSN
  - Squamous cell carcinoma of the skin and oropharyngeal area
  - Short latency
  - Can occur at different localizations
- Association with viral infection
  - HCV infection associated with hepatocellular cancer
  - HPV-related precancer or second cancer, with increased cumulative incidence for cervical, head and neck, vulvar, vaginal, anal, and penile second cancer (Zhao et al. 2021)

47.4 Frequency and Risk Factors (See Table 47.1)

47.4.1 Remarks on SSN

The CI of second solid cancer is 2.2% at 10 years and 6.7% at 15 years (Rizzo et al. 2009).

Increased risk for SSN after HCT has been demonstrated from breast, thyroid, skin, liver, lung, oral cavity and pharynx, brain and CNS, bone and connective tissue cancers and malignant melanoma.

Table 47.1 Frequency and risk factors

Type of SN	Frequency	Risk factors
t-MN	Great variability on the CI of t-MN after auto-HCT • In lymphoma patients between 1% at 2 years up to 24% at 43 months • Lower CI for patients treated for breast cancer, germ cell tumor, and multiple myeloma • Rare n-MN after HCT for AID CI depends mainly on pretransplant cytotoxic and radiation therapy and the use of TBI CI of t-MN after allo-HCT: 0.06–0.67% at 3 years (Yamasaki et al. 2017)	• Quantity of pretransplant chemotherapy (see pathogenesis) and local radiotherapy • Conditioning with TBI • Older age at HCT t-MN are mainly observed after HCT for lymphoma (NHL, HL), probably because lymphoma patients often receive more cytotoxic therapy before HCT The use of lenalidomide (also post-transplant) for multiple myeloma is associated with an increased risk of t-MN among auto-HCT recipients (Rosenberg et al. 2021; Inamoto et al. 2015)
DDM	Rare complication, with a CI <1% at 15 years Possibly underestimated (difficulty to prove donor type of malignant cells) Could represent up to 5% of posttransplant leukemia "relapses"	No clear risk factor defined (too few, heterogeneous DCL) Possible risk factors (Engel et al. 2019) • Malignant donor clone in the transplant • G-CSF therapy • In vivo T cell depletion • Multiple transplantations
SSN		
Breast, thyroid, bone, melanoma, connective tissue, brain, BCC	Breast cancer: 11% at 25 years (Friedman et al. 2008) Thyroid cancer: SIR 3.2 compared to general population (Cohen et al. 2007) BCC: 6.5% at 20 years (Leisenring et al. 2006)	Radiation before HCT or TBI Younger age at radiation Longer follow-up Light-skinned patients (BCC) T-cell depletion (melanoma)

(continued)

Table 47.1 (continued)

Type of SN	Frequency	Risk factors
SSC of skin, oral cavity, and esophagus	SCC of the skin: 3.4 at 20 years (Curtis et al. 2005) Increased risk of atypical melanocytic proliferations and nonmelanoma skin cancer in pediatric HCT recipients (Song et al. 2017)	Chronic GVHD Prolonged GvHD therapy IS including azathioprine Male sex Unrelated with radiation At any time after HCT
Hepatocellular carcinoma	Patients with HCV infection: CI 16% at 20 years (de Latour et al. 2004)	Chronic HCV infection Cirrhosis
Lung cancer	SIR 2.59 after BuCy (Majhail et al. 2011)	Conditioning with Bu-Cy Smoking prior to HCT
Genital and anal cancer	Cervical SIR 2.13; anal SIR 4.85; vulvar and vaginal SIR 22.76; penile SIR 22.76 compared to general population (Zhao et al. 2021)	HPV reactivation (in female and male)

BCC basal cell carcinoma of the skin, *SSC* squamous cell carcinoma, *CI* cumulative incidence, *AID* autoimmune disorders, *SIR* standardized incidence ratio

An individual patient can present several subsequent different SSN after HCT. Up to five different solid cancers have been observed in a patient treated with allo-HCT.

Digestive system and particularly colorectal cancers have not been proven to be increased after HCT (Heydari et al. 2020). In non-transplanted cancer patients, second colorectal cancers are increased when treated with abdominal radiation (Henderson et al. 2012; Rapiti et al. 2008; van Eggermond et al. 2017).

So far there are few long-term data on SSN after RIC. A single-center study shows an increased rate of SSC compared to MAC during the first 10 years post-HCT (Shimoni et al. 2013). The 10-year cumulative incidence of SSC following RIC reached 12.9%, with an overall SIR of 1.03–1.9, mainly significant for head and neck and bladder cancers. Second malignancy seems to be higher than expected in treated allo-HCT recipients conditioned with RIC but still longer follow-up time with larger cohorts is needed for definitive assessment (Del Galy et al. 2022).

The outcome of SSC is mainly dependent on the type of second cancer. Standardized mortality ratio was higher, compared with de novo solid cancers, for melanoma, prostate, breast, kidney, bladder, colorectal, and endometrial cancers but not for the other cancers (Tichelli et al. 2019). Most subsequent solid cancers occurred at younger ages than primary cancers, emphasizing the need for cancer screening at younger ages (Inamoto et al. 2018).

47.5 Screening (Majhail et al. 2012; Inamoto et al. 2015) (See Also Chap. 21)

47.5.1 Therapy-Related Myeloid Neoplasms

Annual monitoring of full peripheral blood counts during the first 10 years after auto-HCT (most t-MN occur within 10 years after HCT).

If unexpected abnormalities in the blood count are observed (increased MCV, cytopenia, dysplasia in peripheral blood, thrombocytosis, leukocytosis, monocytosis), extended analysis of blood and bone marrow is warranted, including cytogenetics and NGS (Bachiashvili et al. 2022).

47.5.2 Donor-Derived Malignancy

Chimerism monitoring of the malignant cells is needed in case of "relapse" or new hematological malignancy after allo-HCT.

Whether to search for clonal hematopoiesis, lymphoid malignant clone, and germline muta-

tions in the donor in case of DDM remains controversial.

47.5.3 Second Solid Cancer (Socie and Rizzo 2012; Inamoto et al. 2015)

Lifelong screening for SSN is recommended after auto-HCT and allo-HCT.

General recommendations are as follows:

- During annual control, clinical screening and reviewing for possible symptoms of SSN.
- Follow country-specific or international guidelines for cancer screening recommendations for the general population.
- Be informed and counselled about the risk of SSN.

Specific recommendations are included in Table 47.2.

Table 47.2 Screening for second solid cancer after HCT

Skin	*All patients* Encouraged to: • Perform regularly genital/testicular and skin self-examination • Avoid unprotected UV skin exposure Skin examination by dermatologist every 1–2 years *Patients at risk* More frequent examination by dermatologist • After first skin cancer • Patients with chronic skin GvHD
Oral cavity and pharynx	*All patients* Examination during annual control *Patients at risk* Annual control by specialist if severe oral and pharynx GvHD Histology in case of suspicious lesion
Thyroid	*All patients* Annual thyroid palpation to identify suspicious thyroid nodules *Patients at risk* (after TBI or local radiation) Regular thyroid ultrasound Fine-needle aspiration in case of a suspicious nodule
Breast	*All patients* Discuss breast self-examination with their physician *Patients at risk* (after TBI or local radiation) Screening mammography every 1–2 years starts at the age of 25 years or 8 years after radiation, whichever occurs later, but not later than age of 40 years
Vagina, Cervix	*All patients* Screening with pap smears every 1–3 years in women older than 21 years old or within 3 years of initial sexual activity, whichever occurs earlier *Patients at risk (HPV positive)* Regular control by specialist of genital organ and anal region
Lung	*All patients* Encouraged to avoid smoking and passive tobacco exposure *Patients at risk* Patients at risk (high-dose busulfan conditioning and smoking), chest CT
Liver	*Patients at risk* Patients with known HCV infection should be assessed for fibrosis/cirrhosis of the liver 8–10 years after HCT (biopsy; fibroscan)
Colorectal	*All patients* Screening should start at age 50 years in the absence of a family history (first-degree relative diagnosed with colorectal cancer before age 60 years): annual fecal occult blood testing, sigmoidoscopy every 5 years, with fecal occult testing every 3 years, or colonoscopy every 10 years
Prostate	*All patients* No specific recommendations

47.6 Treatment

Neoplasm	Treatment
t-MN	Same treatment than de novo myeloid neoplasms Early donor search and rapid allo-HCT (Finke et al. 2016; Kroger et al. 2011; Metafuni et al. 2018) Decision-making including consideration of cumulative toxicity due to previous HCT, age, and comorbidity
DDM	No standard treatment Treatment depends on the nature of disease Reported treatments (Engel et al. 2019) • Retransplantation • Conventional chemotherapy • DLI • Palliation
SSN	Should be treated as de novo cancers of the same type

47.7 Outcome

Neoplasm	Outcome
t-MN	Generally very poor Median survival of 6 m Identical outcome than t-MN in general
DDM	Few data available In most cases, mortality high and OS poor In a small series of 47 DDM, median survival 32.8% months Death mainly due to progression or relapse of DDM
SSN	Mainly dependent on the type of SSN (Ehrhardt et al. 2016; Tichelli et al. 2019; Inamoto et al. 2018) Favorable outcome • Thyroid, breast, prostate, melanoma, cervix Intermediate outcome • Oropharyngeal, colorectal, bladder, renal, ovarian, endometrial Poor outcome • Pancreas, lung, brain, hepatobiliary, esophageal

Key Points

- Three types of secondary neoplasia may occur after HCT: therapy-related myeloid neoplasms (t-MN), mainly after auto-HCT; donor-derived malignancy (DDM) after allo-HCT; and second solid neoplasia (SSN) after auto-HCT and allo-HCT.
- Pretreatment or conditioning with radiation and/or chemotherapy including alkylating agents, anthracyclines, and topoisomerase II inhibitors is mainly responsible for t-MN.
- DDM are extremely rare and are either transmitted from the donor or newly transformed in the host.
- Non-squamous second solid cancers (breast, thyroid, brain, etc.) are strongly related to local radiation or TBI and occur with long delay after HCT. Squamous cell carcinoma of the skin, the oral cavity, and the pharynx is related with chronic GVHD and can occur early after HCT. Conditioning with RIC does not seem to reduce the risk of second cancer.
- Outcome of t-MN is poor, and allogeneic HCT represents the only curative treatment.
- Outcome of SSN depends mainly on the type of second cancer; second solid cancer should be treated as a de novo cancer of the same type.

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Part VI

Specific Organ Complications

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